Drifting by Intention – Four Epistemic traditions in Constructive Design Research

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ABSTRACT

It is assume that to appreciate a knowledge contribution in research-through-design, we all agree on what the act of designing is and should deliver in research. However, just from a glance at contributions in an HCI context, this is far from the case. The course is based on the book: *Drifting by intention – four epistemic traditions in constructive design research* authored by the instructors. It unpacks different ways of knowing in practice-based design and provides operational models and hands-on exercises applied on participants cases to help plan and articulate the contribution of design in each participant's individual research project.

CCS CONCEPTS

• **Human-centered computing, Human computer interaction (HCI), HCI theory, concepts and models**;

KEYWORDS

Research through Design, Constructive Design Research, Drifting, Epistemology

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1 INTRODUCTION

The conference and community of CHI is inherently multi-disciplinary and thus it encompass and welcome numerous epistemologies. This is generally appreciated, but also at times spur frustrations in the community. Design as a creative and intentional practice [7] has manifested itself as a part of the CHI pallet of disciplines over that past 20-some years. However, moving design beyond being a solutions-providing discipline to also encompass design as a research approach under framings such as Research through Design [2][11] or Constructive Design Research [5], have fostered contestation on the type of contribution design is capable of delivering, and how such contributions might be conveyed in papers [3]. Interestingly enough, design, both is bridge, art. engineering, humanities, and science and can be said to be in a somewhat similar situation to CHI where several epistemic traditions co-exist [6].

In maturation of practice-based research a cacophony of academic and artistic voices have risen advocating for increased rigor [12], warning against scientification [4], promoting predictability [8], encouraging activism [1] in design research. And in the past 10 years a wealth of books and handbooks have been published to guide design researcher in producing knowledge e.g. [5], [9], [10].

This course takes point of departure in the 2020 Krogh & Koskinen book "*Drifting by Intention – Four Epistemic Traditions from within Constructive Design Research*"[6]. It is the claim of the book that designers drift, design research drift and so do most other research disciplines. In the context of the book, *drifting* simply means those actions that take design away from its original brief or question, and lead somewhere else. This process can sometimes lead to surprising results, but it does not have to. It is in and through these drifts that knowledge in constructive design research emerge. The modesty and mundaneness of the notion of drifting is also counter to the other (often misleading) dramatic rhetoric's of blue-sky thinking, radical innovations, disruptive innovations, counterintuitive findings, and so forth, often found in design research literature. It may be that constructive design research delivers such breakthroughs, but these heroic moments are rare, and as the book [6] argues, it should not be turned into a standard expectation or a norm. The book and the course take on *drifting* is epistemological. The underlying argument is that how drifting happens depends on how designers understand knowledge they produce: where is its location, what acts create it, how binding it is supposed to be, and ultimately, what drives knowledge creation?

2 BENEFITS AND BACKGROUND OF THE COURSE

As such the course will: 1) Introduce constructive design research as a research field, 2) exemplify and unpack the identified epistemologies in the field and 3) provide tools and models for planning and articulating design as a knowledge producing activity in research and finally 4) unpack the concept of drifting as descriptive for work, experimentation and evaluation in constructive design research.

Beyond the historical and methodological framing of constructive design research, the course will help its participants acquire theoretical insights of constructive design research through hands-on experience using the tools and models presented in the book. This will help the HCI researchers and practitioners position, plan, guide, execute and evaluate constructive design research in both university and industry contexts in and among other research approaches.

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2.1 Background

The book *Drifting by intention – Four Epistemological Traditions from within Constructive Design Research* [2020 Krogh and Koskinen] is based on concepts, models and tools developed for doctoral courses in constructive design research hosted in Nordic countries between 2011 – 2016. The book and its concepts have been presented in numerous invited talks and seminars at design schools (e.g. Goldsmiths, London and Elisava, Barcelona) and departments in universities (e.g. Politecnico di Milano, Dipartemento di Design and Hong Kong PolyU, School of Design) and design conferences (e.g. Cumulus and Design for Transition) in Europe and Asia. Since 2017 the book and its content has served as course material in doctoral education at Jiangnan University, School of Design, China. As a document of this success the book is currently being translated in to Chinese. These widespread activities and interests across cultural and language barriers points to a set of models, tools, theories and conceptions that are relevant to a global audience spanning service design, interaction design product design, engineering and arts' crafts with the shared interest of constructive design research. Communities that are well represented in CHI too.

3 INTENDED AUDIENCE(S)

The key CHI audiences of the course are: master students working to become doctoral students, Doctoral students using design to build knowledge, doctoral student supervisors. Industry researchers in interdisciplinary constructive work environments of both small and large scale companies; supervisors of doctoral students venturing into research-through-design and constructive design research. The key takeaways for each of these audiences are for: Master students insights into how to frame constructive design research; Doctoral students will get acquainted with practical tools for documenting, tracking and discussing research while agilely steering the research work in a fruitful direction; Supervisors will get insights into how other constructive design researchers worked with dilemmas native to the wicked problems of constructive design research. Researchers in industry will find tools and inspiration for how and under which conditions it is fruitful to embed the research approach in their particular context.

4 CONTENT

The course progress from overarching theoretical lectures to introductory lectures on tools and models and interactive exercises providing opportunities for the participants to get hands-on experiences with the tools, models and theoretical concepts presented both in the foundational book [6] and during the lectures. The hands-on interactive parts will take point of departure in the projects and research challenges that the participants bring forward.

The course is planned to run in four modules of 80 min following the structure below:

Module 1 (80 min);

Lecture: Drifting from Frayling to constructive design research and its epistemic traditions

Module 2;

Exercise in break-out groups (total of 45 min): Based upon participant's position papers and discussions four groups will be formed in line with the dominant epistemic tradition (15min). Positioning and discuss each participants research in relation to the epistemic tradition of the group (30 min).

Lecture (35 min): Knowledge-Relevance model and ways of drifting in constructive design research

Module 3;

Exercise individually (40 min): Map a current/ recent constructive design research experiment using the presented tools and models

Exercise in groups according to dominant epistemic tradition (40 min): participants present the mapping exercise.

Module 4;

Exercise (30 min): Short Individual presentations, Group discuss and note similarities and differences revealed through mapping exercise.

Plenum: Group presentations of findings and discussions (40 min); Wrap up by instructors (10 min)

5 PREREQUISITES, PRACTICAL WORK AND ENROLMENT

The course will be advertised through SIGCHI and other relevant mailing lists and social media channels. Interested participants will be asked to submit a 2 pages position paper describing their research and its experimental challenges. It is an advantage if attendees have experience in design as a constructive practice such as product design, interaction design, design engineering, service design and similar.

In line with CHI2021 being a hybrid conference the material and structure of the exercises allow for on-line teaching and hybrid formats too; involving both on-site and on-line participants. The course instructors have experience with teaching formats ranging from fully online co-present and a variety of hybrid formats in between. We will use a suite of (freeware) tools to facilitate: awareness of participants and screen sharing (e.g. zoom or teams); individual and collaborative work on and sharing of material produced during the course (e.g. Mural or Miro) and structured written comments (e.g. Slack). Upon notification of acceptance to the course, participants will be informed about the suite and advised tutorials.

6 INSTRUCTOR BIOGRAPHIES

Peter Gall Krogh is trained as architect and product designer. He is Professor in Design and heads the Socio-Technical Design group at the Department of Engineering at Aarhus University. Prior to this he was professor in design at Aarhus School of Architecture, visiting professor in Politecnico di Milano, Hong Kong PolyU and recently at Jiangnan University.

He contributes to service and interaction design both in doing and theorizing based on co-design techniques with a particular interest in aesthetics, collective action and proxemics. In recent years this has played out in relation to designing for patient experiences in healthcare. His recent book: Drifiting by Intention: Four Epistemic Traditions from within Constructive Design Research describes what design look like and how it can be approached when developing knowledge is equally important as providing opportunities by design. He has published mores than 70 papers, chaired several conferences and held numerous editorial positions in design research and SIGCHI publication fora. He has supervised more than 10 PhDs and examined more than twice as many.

Ilpo Koskinen has a PhD in sociology, but he has worked as a professor of design since 1999 in Helsinki, Melbourne, Hong Kong, and now in Sydney.

His main research interests have been in mobile multimedia, the relationship of design and cities, and interpretive design methodology. Some of his main publications include books Mobile Image, Empathic Design, Mobile Multimedia in Action, Design Research through Practice. From Lab, Field, Showroom, and Drifting by Intention: Four Epistemic Traditions from within Constructive Design Research. His most recent interest have been around social design and unobtrusive design methods. He has published about 150 papers, some of them good (he thinks), supervised to completion 13 PhD students and examined about 40, held numerous editorial positions, and chaired several conferences, most recently DIS 2018.

7 RESOURCES

The foundational book of the course: Drifting by Intention – Four Epistemic Traditions from within Constructive Design Research is available from springer: https://www.springer.com/gp/book/9783030378950 and most research libraries ISBN: 978-3-030-37896-7

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